

Smart AI Medical Report

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Patient Profile

YashM. Patel, Male, 21 years

Summary of Results

Patient profile: YashM. Patel, 21-year-old male. Comprehensive metabolic panel results indicate elevated liver enzymes (ALT) and hypoalbuminemia. Predictive model flags the profile as Abnormal due to discordant liver function tests and low serum albumin.

Detailed Test Explanations

AST (SGOT): 16.00 U/L (normal range: 15-40 U/L). Indicates normal hepatic transaminase activity.

ALT (SGPT): 100.50 U/L (high range: >49 U/L). Elevated ALT suggests hepatocellular injury.

AST:ALT Ratio: 0.50 (low range: <0.90). Subnormal ratio may indicate non-alcoholic fatty liver disease or other hepatic pathologies.

GGT: 10.20 U/L (normal: 0-73 U/L). Normal gamma-glutamyl transferase activity.

ALP: 15.40 U/L (low range: <30 U/L). Hypo-alphaproteinemia suggests reduced cholestasis.

Bilirubin Total: 0.60 mg/dL (normal: 0.30-1.20 mg/dL). Within normal limits.

Bilirubin Direct/Indirect: 0.10 mg/dL (normal: unspecified). Low conjugated bilirubin indicates minimal cholestasis.

Total Protein: 6.39 g/dL (normal: 5.70-8.20 g/dL). Within normal range.

Albumin: 2.00 g/dL (low range: <3.20 g/dL). Hypoalbuminemia indicates synthetic liver dysfunction.

A:G Ratio: 0.10 (low range: <0.90). Abnormal albumin-to-globulin ratio suggests chronic liver disease or nephrotic syndrome.

Overall Interpretation

The patient presents with isolated hypertransaminasemia (ALT 100.50 U/L), hypoalbuminemia (2.00 g/dL), and reduced AST:ALT ratio (0.50). This pattern is atypical for alcoholic hepatitis (where AST typically exceeds ALT) and more consistent with non-alcoholic fatty liver disease or early chronic liver injury. The low A:G ratio further supports chronic liver pathology. Despite the asymptomatic status, the combination of elevated transaminases, hypoalbuminemia, and abnormal ratio necessitates evaluation for advanced fibrosis or hepatocellular carcinoma risk.

Recommendations

1. Diagnostic Workup:

- % Enhanced Liver Fibrosis (ELF) panel (P5P)
- % Hepatitis B/C serology
- % Alpha-fetoprotein (AFP) and Des-gamma carboxyprothrombin

2. Lifestyle Interventions:

- % Weight management if obese
- % Alcohol abstinence
- % Regular exercise and dietary modifications

3. Monitoring:

- % Repeat liver enzymes in 3-6 months
- % Abdominal imaging if fibrosis risk >3%

4. Referral:

- % Gastroenterologist consultation for confirmed chronic liver disease

Disclaimer

This report provides a summary of laboratory findings and is not intended to replace clinical judgment. All results require confirmation and must be interpreted within the context of the patient's complete medical history and physical examination.